SOP-02: Dimethyl Sulfoxide (DMSO)

Roberts Lab, School of Aquatic and Fishery Sciences - Rev. 0 - September 1, 2026

PARTICULARLY HAZARDOUS SUBSTANCE (PHS): NO - Sections 9, 10 and 11 are optional; they are completed below as good practice.

Section 1 - Lab-Specific Information

Building/Room(s) covered by this SOP	Fisheries Teaching & Research (FTR) Building, Room 209
Unit or department	School of Aquatic and Fishery Sciences (SAFS), College of the Environment
Principal Investigator	Steven Roberts (work phone 206-685-3742)
PI signature / date	________________________________ Date: _____________
This SOP was created by (if not PI)	________________________________ Date: _____________
Emergency contact	9-1-1 (campus phone) | EH&S 206-543-7262 (M-F 8-5) | 206-685-UWPD (8973) after hours
Date of last review	_______________ Next review due: _______________ (review annually and after any incident or process change)

Chemicals covered by this SOP

Chemical name (as listed in MyChem)	CAS number
Dimethyl sulfoxide for molecular biology (D8418)	67-68-5
Methyl sulfoxide (AC295520000)	67-68-5

Safety Data Sheets for every chemical listed above must be readily available in the laboratory. SDSs are accessible through MyChem.

Section 2 - Hazards

Applicable GHS pictograms: Flame (combustible liquid, Cat 4); Exclamation mark (irritant)

- Stock chemicals: neat DMSO, 100 mL and 2 L containers. Intermediates and working solutions: DMSO used as a cryoprotectant in cell freezing medium (typically 5-10% v/v in culture medium or seawater), and as a solvent for stock solutions of small molecules such as 5-azacytidine, resazurin, and serotonin. Wastes: neat DMSO residue, DMSO-containing freezing medium, and DMSO stock solutions of dissolved solutes.
- The dominant hazard of DMSO is not its own toxicity but its ability to carry dissolved solutes rapidly through intact skin. Any chemical dissolved in DMSO - including 5-azacytidine, other cytotoxic or reproductive-toxin compounds, and dyes - must be assumed to be dermally absorbed on contact. Treat every DMSO solution as at least as hazardous as the most hazardous chemical dissolved in it.

- Physical hazards: Combustible Liquid Class IIIA (flash point 95 degC); GHS Flammable liquids Category 4. DMSO freezes at 18.5 degC, so bottles solidify in a cold room or a cool laboratory; warm gently in a 37 degC water bath, never with a flame or heat gun. Neat DMSO can react violently with strong oxidizers, acid chlorides and metal halides.
- Routes of exposure: dermal absorption (primary), eye contact, inhalation of vapor or aerosol, ingestion.
- How exposure might occur: pipetting neat DMSO, preparing drug stock solutions, adding DMSO to cell freezing medium, handling cryovials, and glove permeation during prolonged handling.
- Target organs: skin, eyes; systemically, the target organs are those of whatever solute the DMSO is carrying. Neat DMSO may also cause a characteristic garlic-like taste and breath odor within minutes of skin contact, which is a reliable indicator that absorption has occurred.
- Signs and symptoms of exposure: skin redness, itching, burning or scaling; garlic taste or breath odor; headache, dizziness, nausea; eye irritation.
- GHS classification (from MyChem / SDS): Flammable liquids Category 4. Some SDSs additionally list skin and eye irritation and note 'Other Health Hazards - Including Carcinogens' in the MyChem fire class field; this reflects the carrier-solvent concern rather than an established carcinogenicity classification for DMSO itself.
- No OSHA PEL or WISHA PEL has been established for DMSO. DMSO is NOT flagged as a particularly hazardous substance in MyChem; PHS Sections 9-11 are optional and are provided below because DMSO is routinely used to dissolve 5-azacytidine, which IS a PHS.

Section 3 - Engineering Controls and Personal Protective Equipment (PPE)

Engineering controls

- All handling of neat DMSO, and all preparation of DMSO stock solutions of any solute, is performed in the certified chemical fume hood in FTR 209. Any chemical fume hood used must be tested and passed by EH&S.
- Addition of pre-made DMSO freezing medium to cells may be performed in the biological safety cabinet used for cell work, which provides product protection and personnel protection from aerosols.
- Do not warm solidified DMSO on a hot plate or with a heat gun; use a 37 degC water bath or allow it to reach room temperature on the bench.
- The nearest eyewash and safety shower are located in FTR 209 and are inspected and flushed weekly by lab staff.

Hygiene measures

- Avoid contact with skin, eyes, and clothing. Wash hands after removing PPE, before breaks, and immediately after handling the chemical. If any chemical covered by this SOP comes into contact with any PPE, the PPE shall be immediately removed and discarded properly. Any potentially exposed body parts should be washed immediately.
- No eating, drinking, chewing gum, applying cosmetics, or storing food or drink in any area where these chemicals are used or stored.

Skin and body protection

- Chemically compatible laboratory coats that fully extend to the wrist must be worn, appropriately sized and buttoned to their full length. Personnel must also wear full-length pants, or equivalent, and close-toe shoes. The area of skin between the shoe and ankle must not be exposed.
- Lab coat type: standard barrier lab coat. A disposable chemically resistant sleeve set is worn when handling more than 100 mL of neat DMSO or any DMSO solution of a cytotoxic or reproductive-toxin solute.
- Because DMSO is toxic by skin absorption and enhances the absorption of its solutes, additional protective clothing (chemically resistant apron and disposable sleeves) is required where splashes or skin contact is foreseeable.

Hand protection

- Hand protection IS required for the activities described in this SOP, and glove selection is critical.
- Standard disposable nitrile gloves are NOT adequate for DMSO - DMSO permeates thin nitrile within minutes. Required gloves: butyl rubber gloves (e.g., North B174, Ansell AlphaTec Butyl) for handling neat DMSO, OR double thick nitrile (minimum 8 mil, e.g., Microflex MidKnight XTRA or Halyard Purple Nitrile-Xtra) changed immediately on any contact and at least every 15 minutes for brief, incidental-contact pipetting only.
- Consult your preferred glove manufacturer to ensure that the gloves you plan to use are compatible with DMSO. Gloves must be inspected prior to use, including a check for pinholes.
- Any glove that contacts DMSO is removed and replaced immediately - do not continue working in it.
- Use proper glove removal technique (without touching the glove's outer surface). Dispose of contaminated gloves after use in accordance with applicable laws and good laboratory practices. Wash and dry hands immediately after glove removal.

Eye protection

- ANSI Z87.1-compliant eye protection IS required for all work with DMSO. Ordinary prescription glasses will NOT provide adequate protection unless they also meet the Z87.1 standard and have compliant side shields.
- Minimum eye protection: ANSI Z87.1 safety glasses with side shields for microliter pipetting inside the fume hood; chemical splash goggles for any transfer of more than 10 mL of neat DMSO or any DMSO solution containing a cytotoxic solute.

Respiratory protection

- Respiratory protection is NOT required for the activities described in this SOP.
- DMSO may be used outside of the chemical fume hood only for closed-tube operations (adding pre-made freezing medium to capped cryovials, transferring capped vials). All open handling of neat DMSO and all preparation of DMSO solutions must be done in the fume hood.
- Respirators should be used as a last line of defense (i.e., after engineering and administrative controls have been exhausted), and when any Action Limit or Occupational Exposure Limit has been exceeded. If a potential exposure hazard cannot be eliminated, contact the EH&S Respiratory Protection Program administrator.

Section 4 - Special Handling and Storage Requirements

- Quantity limits: no more than 100 mL of neat DMSO is dispensed at one time; drug stock solutions are prepared at the smallest practical volume (typically 0.5-5 mL).
- Practices beyond general laboratory rules: never handle DMSO with bare hands or with gloves that have contacted any other chemical; assume any spilled DMSO on a glove has already carried that chemical to the skin. Never wear jewelry or a watch that could trap DMSO against skin.
- Every DMSO stock solution container and cryovial must be labeled with the full name of the dissolved solute, the concentration, the solvent (DMSO), the date and the preparer's initials - including tubes stored at -20 degC or -80 degC.
- Storage: DMSO is stored in the chemical storage cabinet in FTR 209 at room temperature, upright, in secondary containment, away from strong oxidizers (copper(II) nitrate), acid chlorides, and strong acids (sulfuric acid, hydrochloric acid). DMSO solutions of drugs are stored in a labeled, closed secondary container in the -20 degC freezer.
- Transport: carry containers larger than 100 mL in a bottle carrier; transport cryovials in a closed rack or secondary container.
- Unattended operations: none are performed with DMSO.
- Clean the chemical fume hood upon completion of tasks with 70% ethanol followed by deionized water.

- Clean all contaminated surfaces with soap and water followed by 70% ethanol and dry.
- Place all contaminated disposable items in appropriate laboratory waste for disposal.
- Non-disposable/re-usable utensils, glassware, and other surfaces contaminated with DMSO must be decontaminated at the end of the laboratory work session. Complete this inside the chemical fume hood before removing any of the items.
- When work is completed, remove gloves and wash hands with soap and water.

Section 5 - Spill and Accident Procedures

Spill response

- Chemical spills must be cleaned up as soon as possible by properly protected and trained personnel. All other persons should leave the area. Do not attempt to clean up any spill if not trained or comfortable.
- Small spill - less than 50 mL of neat DMSO inside the fume hood, containing no hazardous solute: may be handled by trained lab staff. Wearing goggles, butyl gloves and a lab coat, absorb with universal absorbent pads from the laboratory spill kit, working from the outside in.
- Any spill of a DMSO solution containing 5-azacytidine or another cytotoxic, carcinogenic or reproductive-toxin solute, and any DMSO spill greater than 50 mL or outside the fume hood: treat as a high-hazard spill. Restrict access, post a sign, and call EH&S at 206-543-0467 (business hours) or 9-1-1.
- Do not use paper towels alone to wipe a DMSO spill - wiping spreads a skin-permeant liquid across a wide area and increases contact risk. Absorb first, then decontaminate.
- Liquid spills only are anticipated; DMSO is not handled as a solid other than as a frozen bottle, which is warmed intact.
- PPE required for cleanup: chemical splash goggles, face shield, butyl rubber gloves over nitrile, buttoned lab coat with disposable chemically resistant sleeves or apron, full-length pants and close-toe shoes.
- Spill area must be cleaned up in the following manner: after absorbent removal, wash the area thoroughly with detergent solution followed by clean water, then wipe with 70% ethanol and dry.
- Spill cleanup materials must be disposed of in the following manner: double bag all absorbent and contaminated disposables in plastic bags, place in a screw-top container, label with a UW Hazardous Waste Label listing 'DMSO' and any solute, and submit a collection request to EH&S.
- For questions on spill cleanup, contact EH&S spill consultants at 206-543-0467 during normal business hours.
- Any spill, exposure or near miss incident requires the involved person or supervisor to complete and submit an incident report on the EH&S website within 24 hours.

Exposure procedures - first aid

- Perform first aid immediately. Refer to the SDS for additional chemical-specific guidance.
- Skin exposure: this is the exposure of greatest concern. Immediately remove contaminated gloves and clothing and flush the skin with copious running water for at least 15 minutes using the nearest sink or safety shower. Do NOT use solvents or scrub. Report the identity of any solute dissolved in the DMSO to the treating clinician, because the solute may have been absorbed systemically.
- Eye exposure: use the eyewash for 15 minutes while holding the eyelids open.
- Inhalation exposure: move out of the contaminated area to fresh air; get medical help.
- Ingestion: rinse mouth with water; do not induce vomiting; call 9-1-1.
- Sharps injury (needle stick or subcutaneous exposure): scrub the exposed area thoroughly for 15 minutes using warm water and sudsing soap.

- Get help. Call 9-1-1 or go to the nearest Emergency Department; provide details of exposure: agent (state both DMSO and the solute), dose, route of exposure, and time since exposure. Bring the SDS and this SOP to the Emergency Department.
- Notify your supervisor as soon as possible. Secure the area before leaving.
- Notify EH&S immediately after providing first aid. During business hours (M-F, 8-5) call 206-543-7262; outside business hours call 206-685-UWPD (8973).

Section 6 - Waste Accumulation and Disposal Procedures

- Waste streams: (a) neat DMSO; (b) DMSO solutions of non-hazardous solutes; (c) DMSO solutions of cytotoxic, carcinogenic or reproductive-toxin solutes (e.g., 5-azacytidine) - this stream is managed under SOP-10 as well; (d) DMSO-contaminated solid debris.
- Do NOT pour DMSO down the drain, even dilute. Do not mix DMSO waste with strong acid or oxidizer waste.
- Accumulate liquid waste at the point of generation in a sturdy, chemically compatible HDPE container with a securely closable screw-top lid, kept closed and in secondary containment.
- All chemical waste containers must be labeled with a UW Hazardous Waste Label listing DMSO and every solute with approximate percentage. Refer to 'How to Label Chemical Waste Containers'.
- Solid debris (pipette tips, gloves, wipes) contaminated with neat DMSO or with DMSO solutions of hazardous solutes is collected in a lined, closed, labeled solid hazardous waste container - not the regular trash.
- Manage chemical and hazardous chemical waste separately from other waste streams such as biohazardous waste. Never autoclave chemical waste.
- Cryovials containing DMSO freezing medium that are no longer needed are thawed inside the fume hood and the liquid added to the DMSO liquid waste container; the empty vials are then discarded as contaminated solid waste (or as biohazardous waste if they contained cells).
- To request a collection of chemical waste, submit a form on the Chemical Waste Disposal webpage on the EH&S website or directly in MyChem inventory. Contact EH&S at 206-616-5835 or chmwaste@uw.edu with questions.
- Decontamination of equipment, glassware and controlled areas: wash with detergent and water, rinse three times with deionized water, then wipe with 70% ethanol. Update the MyChem inventory when containers are emptied or received.
- Visit the Hazardous Material Disposal and Recycling webpage on the EH&S website for information on disposing, recycling and surplusing materials.

Section 7 - Protocol

- Procedure A - Preparation of a small-molecule stock solution in DMSO. In the certified fume hood, wearing butyl gloves, goggles and a lab coat, weigh the solid compound (see the relevant compound SOP - for 5-azacytidine, see SOP-10) and add the calculated volume of neat DMSO with a positive-displacement or air-displacement pipette. Cap and vortex to dissolve. Label the tube with compound name, concentration, solvent, date and initials. Aliquot into single-use volumes to avoid repeated freeze-thaw and to minimize repeated handling. Store at -20 degC in a labeled closed secondary container.
- Procedure B - Preparation of cell freezing medium. In the biological safety cabinet or fume hood, add neat DMSO to chilled culture medium or filtered seawater to a final concentration of 5-10% v/v, mixing continuously. DMSO dilution is exothermic; add DMSO to medium, never medium to DMSO, and keep the mixture on ice. Label the bottle and use within the interval specified in the protocol.

- Procedure C - Cryopreservation. Add freezing medium to the cell pellet, transfer to labeled cryovials, cap securely, and place in a controlled-rate freezing container at -80 degC. Wear cryogenic gloves over nitrile when handling vials removed from -80 degC storage; a vial that has warmed can pressurize.
- Refer to Section 2 of the UW Laboratory Safety Manual on the EH&S website for additional guidance on chemical management and preparation for use for particularly hazardous substances (PHSs).
- NOTE: Any deviation from this SOP requires approval from the Principal Investigator.

Section 8 - Special Precautions for Animal Use

No. N/A - DMSO is not administered to animals under this SOP. If DMSO is ever used as a vehicle for exposing live animals (including bivalves or finfish) to a test compound, that use must be documented and approved by IACUC and this SOP must be revised before the work begins.

IACUC protocol number (if applicable): ______________________________

PARTICULARLY HAZARDOUS SUBSTANCE (PHS): NO - Sections 9, 10 and 11 are optional; they are completed below as good practice.

Section 9 - Approvals Required

- All staff working with DMSO must be trained on this SOP prior to starting work. They must also review the DMSO SDS, and it must be readily available in the laboratory. All training must be documented and maintained by the PI or their designee.
- Prerequisite training: EH&S Laboratory Safety Training and EH&S Managing Laboratory Chemical Waste training.
- A worker must have documented, PI-approved training on glove selection and DMSO skin-absorption hazards, including one supervised preparation of a DMSO stock solution, before performing Procedure A independently for the first time.
- Preparing a DMSO stock solution of any cytotoxic, carcinogenic or reproductive-toxin compound (including 5-azacytidine) requires specific written authorization from the PI and completion of the training required by SOP-10.
- No medical examination or respirator authorization is required for the work described.

Section 10 - Decontamination

- Equipment: wipe pipettes, racks and balance surfaces with soap and water, then 70% ethanol, and allow to dry. Weigh boats and spatulas used with DMSO solutions of hazardous solutes are discarded as hazardous solid waste rather than reused.
- Glassware: rinse into the DMSO waste container, then wash with detergent and water and rinse three times with deionized water.
- Fume hood: wipe the work surface with detergent solution, then 70% ethanol, at the end of each work session. Use a disposable bench liner under DMSO work and discard it as hazardous solid waste.
- Controlled areas: there are no glove boxes or restricted-access hoods in FTR 209; the DMSO designated area is decontaminated as described above.

Section 11 - Designated Area

- All open handling of neat DMSO and all preparation of DMSO stock solutions: the certified chemical fume hood in FTR 209, on a disposable bench liner.
- Addition of pre-made DMSO freezing medium to cells and handling of capped cryovials: the biological safety cabinet and the adjacent cell-culture bench in FTR 209.
- Storage: chemical storage cabinet, FTR 209 (neat DMSO); labeled closed secondary container in the -20 degC freezer (DMSO stock solutions).

Section 12 - Documentation of Training

- Prior to using substances included in this SOP, laboratory personnel must be trained on the hazards described in this SOP, how to protect themselves from the hazards, and emergency procedures.
- Ready access to this SOP and to a Safety Data Sheet for each hazardous material described in the SOP must be made available in the lab space(s) where these substances are used.
- The Principal Investigator, or Responsible Party if the activity does not involve a PI, must ensure that their laboratory personnel have attended appropriate laboratory safety training (and refresher training where applicable).
- Training must be repeated following any revision to the content of this SOP.
- Training must be documented. This training sheet is provided as one option; other forms of training documentation (including electronic) are acceptable but records must be accessible and immediately available upon request.

I have read and understand the content of this SOP:

Name (print)	Signature	Date